

SAVITHA VIJAYARANGAN

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ML/AI engineer and M.S. candidate in Applied Data Intelligence at San José State University, taking models from development to deployment across edge AI, NLP and production ML services, built on nearly four years of data and backend engineering.

EXPERIENCE

IDX Exchange · *NLP Engineer Intern* Remote, US · Sep 2026 – Present

- Delivered sub-100ms semantic search over 10,000+ listings with sentence-transformer embeddings and FAISS, evaluated against BM25 and served through an 8-endpoint FastAPI service with caching and rate limiting.
- Reached 87% F1 on information extraction and 92% query-parsing accuracy with a named-entity extractor and an object-oriented natural-language query parser that generates parameterized SQL with schema validation.
- Built a Fair Housing compliance checker with 100% recall and 84% precision on labeled violations, and a 200+ term real-estate taxonomy from n-gram analysis of agent listing remarks.

San José State University · *Instructional Student Assistant, Deep Learning (DATA 255) & Generative AI (DATA 266)* San Jose, CA · Jul 2026 – Present

- Developed and refined course curriculum and mentored 100+ graduate students on model deployment and distributed training.
- Automated grading pipelines and submission-management scripts for high-enrollment graduate courses.

HexasenseAI · *ML Engineer Intern* Austin, TX (Remote) · Jun 2025 – Aug 2025

- Deployed optimized ML pipelines that reduced inference latency by 30%, improving real-time system responsiveness.
- Built clinical risk-prediction models on healthcare data, applying Responsible AI practices for fairness, transparency and reliability.

Larsen & Toubro Infotech (LTI) · *System Consultant (Data & Analytics Engineer); Software Engineer* India · Sep 2017 – May 2021

- Built Python, SQL and Airflow ETL pipelines handling 1M+ records/day, integrated 5+ sources into a Snowflake warehouse, and cut pipeline latency by 25%.
- Built Python backend services handling 50K+ transactions/day and delivered 10+ production features via REST APIs in Agile sprints.

PROJECTS

Wildfire Edge Sentinel · HP Edge AI SJSU Hackathon · YOLO11, Qwen2.5-VL, LoRA, ONNX, Docker Sep 2026

- Built offline-first wildfire smoke detection on an HP ZGX Nano: a fine-tuned YOLO11 detector plus a Qwen2.5-VL-7B LoRA distilled from a 32B teacher; the pipeline keeps detecting and queues alerts during network outages.
- Raised detector mAP50 from 0.002 to 0.748 on D-Fire and from 0.198 to 0.718 on lookout-tower smoke; raised VLM source-type agreement from 45.0% to 73.6% on 500 held-out crops.
- Designed edge-vs-cloud escalation that keeps video on-device: on a 266-frame, 6-tower run, cut VLM calls from 266 to 11, tokens from 447K to 3K and video upload from 23 MB to 0 B.

Kayak Travel Simulation System · FastAPI, Kafka, LangChain, OpenAI API, MySQL, MongoDB, Redis, React

- Engineered 3-tier microservices with FastAPI and Kafka for real-time booking feeds, on a MySQL, MongoDB and Redis data layer deployed with Docker.
- Built an AI chatbot with LangChain and the OpenAI API to identify user intent, improving response accuracy by 25%.

GPT-Style Language Model from Scratch · PyTorch, Transformers, Attention Feb – Mar 2026

- Built and trained a ~4.8M-parameter decoder-only Transformer from scratch (no pretrained embeddings) on TinyStories, reaching 0.6438 validation loss; documented failure modes across sampling temperatures.

Image Classification & Super-Resolution on NPU · PyTorch, CNN, ONNX, NPU deployment

- Trained a custom CNN for multi-class ImageNet classification and a super-resolution DNN, and deployed both for real-time inference on NPU hardware by converting PyTorch models to ONNX and compiling to MXQ.

EDUCATION

San José State University · M.S., Applied Data Intelligence · GPA 3.53 · Coursework: ML, Deep Learning, Generative AI, Distributed Systems, Big Data 2025 – Present

B.S. Abdur Rahman University · B.Tech., Electronics & Communication Engineering, 2013 – 2017 · **Harvard Business School Online** · Business Analytics Certification, 2023

SKILLS

ML / DL: PyTorch, Scikit-Learn, TensorFlow (coursework), ONNX, CNNs, Transformers

GenAI / NLP: LLMs, RAG, LangChain, OpenAI API, LoRA/QLoRA, vector databases, FAISS, sentence-transformers, NLTK

Deployment & data: FastAPI, Docker, MLflow, ONNX/NPU deployment, AWS S3, PySpark, Kafka, Airflow, dbt, Snowflake

Languages & databases: Python, SQL, PostgreSQL, MySQL, MongoDB, Redis